Nhan Tran

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EDUCATION

Iowa State University — Ames, IA

Master of Science, Computer Science (GPA 3.8 / 4.0) — December 2025

Bachelor of Science, Computer Science (GPA 3.7 / 4.0) — May 2023

PROGRAMMING SKILLS

- **Languages**: C/C++ • C# • Java • Python • SQL • JavaScript
- **Softwares**: VS Code • Godot • Android Studio • Github • Git • Jira • AWS
- **Libraries**: Emscripten • WASM • Pytorch • Spring Boot

WORK EXPERIENCE

Principal Financial — Ames, IA

Software Developer Intern — June 2022 – August 2023

- Worked in an Agile environment among a team of software engineers.
- Developed and tested frontend and backend of various websites using Java and Spring Boot.
- Processed and visulized financial data using Python and SQL.
- Set up automatic gather, process, and visualize Energy Star certified housing data with AWS Athena, S3, DynamoDB.

Promax Unlimited — Bettendorf, IA

Software Developer Intern — January 2020 – June 2020

- Developed an authentication system for company's app using JavaScript and React Native.

PROJECTS

Procedural Generation Tile Constraints — April 2025

- Demo: pgtc.tranhqnhan.org
- Generated large textures, images from samples with locally self-similar structures.
- Implemented Model Synthesis / WFC on a 2D grid for procedurally generating textures and images from example.
- Mutithreaded application to render the image and computing constraints in parallel.
- Developed with C++, Raylib, Emscripten, WASM, Multithreading.

Quadtree A* Search — March 2025

- Demo: qtas.tranhqnhan.org
- Improved speed of A* path planning on large grids by subdivide the grid with quadtree.
- Implemented quadtree algorithm with constant time neighbor finding to generate region graph.
- Developed with C++, Raylib, Emscripten, WASM.

Multiprocessing Task Scheduler with Ant Colony Optimization — November 2025

- Implemented Ant Colony Optimization algorithm for task scheduling in heterogeneous multiprocessing environment.
- Project for Principles of Operating System class.
- Developed with C and visualized with Raylib.

Iterated Dijkstra Propagation for Robotics Path Planning — November 2025

- Implemented Iterated Dijkstra Propagation for multiobjective path planning.
- Created a multicost resource pool architecture to manage memory efficiently.
- Developed an edge retrieval optimization that speed up the algorithm by seven percent.
- Developed using C++ and tested on the Husarion 2R robot with ROS platform.

AWARDS AND CERTIFICATIONS

Python Business Professionals of America Award — May 2022

Second place in Python programming competition hosted by Business Professionals of America.

Java IT Specialist — May 2022

Certified by Pearson Credly at Business Professionals of America event.

Python IT Specialist — May 2022

Certified by Pearson Credly at Business Professionals of America event.

Iowa Governor Scholar — May 2020

Awarded to the two highest academically performing seniors.

Hawkeye Programming Challenge — April 2019

Second place in programming competition hosted by University of Iowa